

Resume Generator API

What I Understood

The purpose of this project was to understand how a frontend application communicates with a backend service to generate and display dynamic content. The project simulates a Resume Generator where users enter their personal, educational, and professional details through a web interface, and the application generates a formatted resume.

Technologies Used

Frontend

- HTML
- CSS
- JavaScript

Backend

- FastAPI (Python)

Communication

- REST API
- JSON

Project Workflow

Step 1: User Enters Information

The user fills in details such as:

- Full Name
- Email Address
- Phone Number
- Skills
- Education
- Work Experience

Step 2: Frontend Sends Data

JavaScript collects the entered information and sends it to the FastAPI backend using a POST request.

Step 3: Backend Processes Request

The FastAPI backend receives the data through the API endpoint and organizes the information into a resume structure.

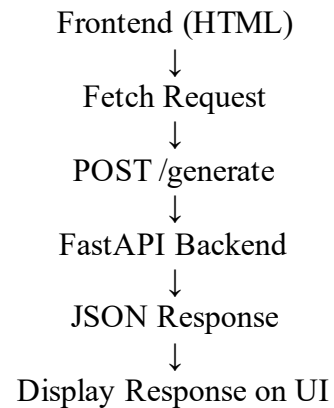
Step 4: Generate Resume

The backend generates a formatted resume response and returns it as JSON.

Step 5: Display Resume

The frontend receives the response and dynamically displays the generated resume to the user.

Flow I Implemented



Screenshots

Frontend Integration

```
... <> index1.html main5.py resume.py <> resume1.html x http://localhost:8000/
backend > <> resume1.html > html > body > script > generateResume > response
*
U
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133     email: document.getElementById("email").value,
134     phone: document.getElementById("phone").value,
135     skills: document.getElementById("skills").value
136             .split(",")
137             .map(skill => skill.trim()),
138     education: document.getElementById("education").value,
139     experience: document.getElementById("experience").value
140 };
141
142 try {
143     const response = await fetch(
144         "http://127.0.0.1:8000/generate-resume",
145         {
146             method: "POST",
147             headers: {
148                 "Content-Type": "application/json"
149             },
150             body: JSON.stringify(data)
151         }
152     );
153
154     const result = await response.json();
155
156     document.getElementById("result").innerText =
157         result.resume || JSON.stringify(result, null, 2);
158
159 } catch(error) {
160     document.getElementById("result").innerText =
161         "Error connecting to API";
162     console.error(error);
163 }
164 }
165 </script>
166
167 </body>
168 </html>
```


<> index1.html main5.py resume.py x <> resume1.html http://localhost:8000/

backend > resume.py > generate_resume

```
def generate_resume(data: Resume):

26     resume = f"""
27     =====
28     |         |         |         |         |         |
29     |         |         |         |         |         |
30     =====
31     📧 Email : {data.email}
32     📞 Phone : {data.phone}
33
34     -----
35     PROFESSIONAL SUMMARY
36     -----
37     Motivated professional with expertise in
38     {', '.join(data.skills)} and a strong interest in
39     continuous learning and career growth.
40
41     -----
42     EDUCATION
43     -----
44     {data.education}
45
46     -----
47     WORK EXPERIENCE
48     -----
49     {data.experience}
50
51     -----
52     TECHNICAL SKILLS
53     -----
54     """
55
56     for skill in data.skills:
57         resume += f"\n• {skill}"
58
59     resume += """
60
61     -----
62     DECLARATION
63     -----
64     I hereby declare that the above information is
65     true and correct to the best of my knowledge.
66
67     Signature
68     -----
69     """
70
71     return {
72         "message": "Resume Generated Successfully",
73         "resume": resume
74     }
```

API Endpoint Testing using FastAPI Swagger UI

FastAPI 0.1.0 OAS 3.1
[/openapi.json](#)

default ^

POST /generate-resume Generate Resume ^

Parameters

Cancel Reset

No parameters

Request body required

application/json v

Edit Value | Schema

```
{
  "name": "Pragati",
  "email": "pragati123@gmail.com",
  "phone": "6362616465",
  "skills": [
    "HTML", "CSS", "Cyber Security"
  ],
  "education": "SSLC , Diploma",
  "experience": "Fresher"
}
```

Execute Clear

Responses

API Response

Responses

Curl

```
curl -X 'POST' \
  'http://localhost:8000/generate-resume' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "name": "Pragati",
    "email": "pragati123@gmail.com",
    "phone": "6362616465",
    "skills": [
      "HTML", "CSS", "Cyber Security"
    ],
    "education": "SSLC, Diploma",
    "experience": "Fresher"
  }'
```

Request URL

<http://localhost:8000/generate-resume>

Server response

Code	Details
------	---------

200

Response body

```

"message": "Resume Generated Successfully",
"resume": {
  "name": "PRAGATI",
  "email": "pragati123@gmail.com",
  "phone": "6362616465",
  "summary": "Motivated professional with expertise in HTML, CSS, Cyber Security and a strong interest in continuous learning and career growth.",
  "education": "BSC in Information Technology",
  "experience": "Fresher",
  "skills": "HTML, CSS, Cyber Security",
  "declaration": "I hereby declare that the above information is true and correct to the best of my knowledge."
}

```

Response headers

```
access-control-allow-credentials: true
access-control-allow-origin: http://localhost:8000
content-length: 1183
content-type: application/json
date: Fri, 12 Jun 2026 07:48:39 GMT
server: uvicorn
vary: Origin
```

Responses

Code	Description	Links
200	Successful Response	<i>No links</i>

Media type

application/json

Controls Accept header.

Example Value	Schema
1	1
2	2
3	3
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99	99
100	100

"string"

Frontend User Interface

Resume Generator

Name

Email

Phone

Skills (comma separated)

Education

Experience

Successful Backend Execution (terminal)

Generated Resume

=====

PRAGATI

=====

Email : pragati123@gmail.com
Phone : 6362616465

PROFESSIONAL SUMMARY

Motivated professional with expertise in HTML, CSS, Cyber Security and a strong interest in continuous learning and career growth.

EDUCATION

SSLC , Diploma

WORK EXPERIENCE

Fresher

TECHNICAL SKILLS

- HTML
- CSS
- Cyber Security

DECLARATION

I hereby declare that the above information is true and correct to the best of my knowledge.

Signature _____

Learning

From this task I learned:

- What an API endpoint is
- How FastAPI works
- How frontend and backend communicate
- How to test APIs using Swagger UI
- How request/response flow works